



SYSTEMS ANALYSIS AND DESIGN

IS104





PROJECT MANAGEMENT

WEEK 5 & 6



Introduction

The Integrated Process Project Management Process is described in detail in this study.

The first step is to choose a project that will add value to the company and create a plan an application containing basic information about the proposed scheme.

Third, the project manager determines the project performance and the tasks to be completed.

Finally, the manager determines the tools, standards, and procedures to be used, as well as opportunities for reuse, how the current project fits into the portfolio of currently developing projects, and the likelihood of reviving the entire structure of the company's portfolio of currently used systems.

Second, analysts behave analysis to determine program performance, economic, and organizational performance; if appropriate, the plan is selected, and the development project is in progress.

Fourth, this project has managers.

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PROJECT

- *In the broadest sense, a project is a specific, finite task to be accomplished. Any activity that results in a deliverable or a product.*
- *Projects always begin with a problem. The project is to provide the solution to this problem.*
- *When the project is finished it must be evaluated to determine whether it satisfies the objectives and goals.*

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What is Management?

Management can be defined as all activities and tasks undertaken by one or more persons for the purpose of planning and controlling the activities of others in order to achieve objectives or complete an activity that could not be achieved by others acting independently.

- *Management functions can be categorized as*
 - *Planning*
 - *Organizing*
 - *Staffing*
 - *Directing*
 - *Controlling*

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What is Project Management?

“The application of knowledge, skills, tools and techniques to project activities in order to meet project requirements”

Project management is a system of:

- *management procedures,*
- *practices,*
- *technologies,*
- *skills, and*
- *experience that are necessary to successfully manage a project.*

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Software Project Management

Concerned with activities involved in ensuring that software is delivered:

- *on time*
- *on schedule*
- *in accordance with the requirements of the organization developing and procuring the software*

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Project Stakeholders

Stakeholders are the people involved in or affected by the project activities

Stakeholders includes:

- ☐ *The project sponsor and project team*
- ☐ *Support staff*
- ☐ *Customers*
- ☐ *Users*
- ☐ *Suppliers*
- ☐ *Opponents to the project*

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Project Characteristics

One clear objective

- *A well-defined set of end results*
- *Goal oriented*
- *End product or service must result*

Finite

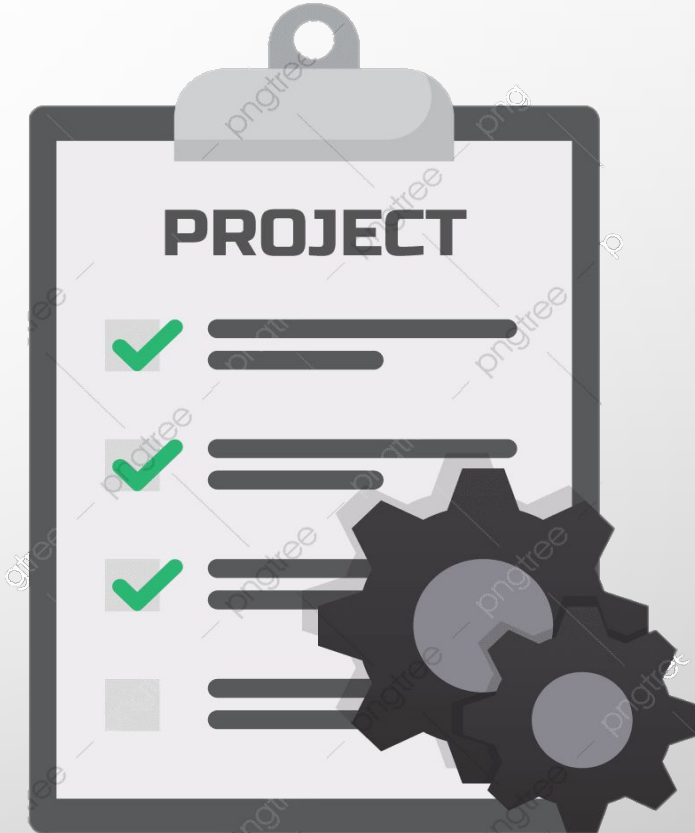
- *Fixed timeline, start date, end date, milestone dates*

Limited

- *Budget, Resources, Time*

Life Cycle

- *Recognizable sequence of phases*



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Project Identification and Initiation

- *Projects are driven by business needs*
 - *Identified by business people*
 - *Identified by IT people*
 - *(better yet) identified jointly by business and IT*
- *The project sponsor believes in the system and wants to see it succeed*
 - *Normally this is a business person*
 - *Should have the authority to move it forward*

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Business Value

Tangible Value

- ✓ *Can be quantified and measured easily*
- ✓ *Example: 2 percent reduction in operating costs*

Intangible Value

- ✓ *Results from an intuitive belief that the system provides important, but hard-to-measure, benefits to the organization.*
- ✓ *Example: improved customer service*

TIME IS MONEY



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Elements of a System Request

Project sponsor

- ✓ *Primary point of contact for the project*

Business need

- ✓ *Reason prompting the project*

Business requirements

- ✓ *Business capabilities the system will need to have*

Business value

- ✓ *Benefits the organization can expect from the project*

Special issues

- ✓ *Anything else that should be considered*



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Feasibility Analysis

Guides the organization in determining whether to proceed with a project.

Identifies the project's risks that must be addressed if the project is approved.

Major components:

- ✓ *Technical feasibility*
- ✓ *Economic feasibility*
- ✓ *Organizational feasibility*

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Feasibility Analysis

A. *Technical Feasibility (Can we build it???)*

Familiarity with application

- *Less familiarity generates more risk*

Familiarity with technology

- *Less familiarity generates more risk*

Project size

- *Large projects have more risk*

Compatibility

- *Difficult integration increases the risk*

B. *Economic Feasibility (Should we build it??)*

- ✓ *Development costs*
- ✓ *Annual operating costs*
- ✓ *Annual benefits (cost savings and revenues)*
- ✓ *Intangible costs and benefits*

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Cost-Benefit Analysis

Performance Measure	Formula
Return on Investment (ROI)	$\frac{\text{Total benefits} - \text{total costs}}{\text{Total costs}}$
Break-Even Point	$\frac{\text{Yearly Net Cash Flow} - \text{Cumulative Net Cash Flow}}{\text{Yearly Net Cash Flow}}$ Use the yearly net cash flow amount from the first year in which the project has a positive cash flow. Add the above amount to the year in which the project has a positive cash flow minus one.
Present Value (PV)	$\frac{\text{Cash flow amount}}{(1 + \text{interest rate})^n}$ n = number of years in the future
Net Present Value (NPV)	$\Sigma \text{ PV Benefits} - \Sigma \text{ PV Costs}$

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Steps for Conducting Economic Feasibility

1. Identifying Costs and Benefits

List the tangible costs and benefits for the project. Include both one-time and recurring costs.

2. Assigning Values to Costs and Benefits

Work with business users and IT professionals to create numbers for each of the costs and benefits. Even intangibles should be valued if at all possible.

3. Determining Cash Flow

Project what the costs and benefits will be over a period of time, usually three to five years. Apply a growth rate to the numbers, if necessary.

4. Determining Net Present Value (NPV)

Calculate what the value of future costs and benefits are if measured by today's standards. You will need to select a rate of growth to apply the NPV formula.

5. Determining Return on Investment (ROI)

Calculate how much money the organization will receive in return for the investment it will make using the ROI formula.

6. Determining the Break-Even Point

Find the first year in which the system has greater benefits than costs. Apply the break-even formula using figures from that year. This will help you understand how long it will take before the system creates real value for the organization.

7. Graphing the Break-Even Point

Plot the yearly costs and benefits on a line graph. The point at which the lines cross is the break-even point.

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Steps for Conducting Economic Feasibility

1. *Identifying Costs and Benefits* The first task when developing an economic feasibility analysis is to identify the kinds of costs and benefits the system will have and list them along the left -hand column of a spreadsheet.

Development Costs	Operational Costs
Development Team Salaries	Software Upgrades
Consultant Fees	Software Licensing Fees
Development Training	Hardware Repairs
Hardware and Software	Hardware Upgrades
Vendor Installation	Operational Team Salaries
Office Space and Equipment	Communications Charges
Data Conversion Costs	User Training
Tangible Benefits	Intangible Benefits
Increased Sales	Increased Market Share
Reductions in Staff	Increased Brand Recognition
Reductions in Inventory	Higher Quality Products
Reductions in IT Costs	Improved Customer Service
Better Supplier Prices	Better Supplier Relations

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Steps for Conducting Economic Feasibility

2. *Assigning Value to Cost Benefits*

Benefits^a	
Increased sales	500,000
Improved customer service ^b	70,000
Reduced inventory costs	68,000
Total benefits	638,000
Development costs	
2 servers @ \$125,000	250,000
Printer	100,000
Software licenses	34,825
Server software	10,945
Development labor	1,236,525
Total development costs	1,632,295
Operational costs	
Hardware	54,000
Software	20,000
Operational labor	111,788
Total operational costs	185,788
Total costs	1,818,083

^a An important yet intangible benefit will be the ability to offer services that our competitors currently offer.

^b Customer service numbers have been based on reduced costs for customer complaint phone calls.

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Steps for Conducting Economic Feasibility

3. Determining the cash flow

	2015	2016	2017	2018	2019	Total
Increased sales	500,000	530,000	561,800	595,508	631,238	
Reduction in customer complaint calls	70,000	70,000	70,000	70,000	70,000	
Reduced inventory costs	68,000	68,000	68,000	68,000	68,000	
TOTAL BENEFITS:	638,000	668,000	699,800	733,508	769,238	
PV OF BENEFITS:	619,417	629,654	640,416	651,712	663,552	3,204,752
PV OF ALL BENEFITS:	619,417	1,249,072	1,889,488	2,541,200	3,204,752	
2 Servers @ \$125,000	250,000	0	0	0	0	
Printer	100,000	0	0	0	0	
Software licenses	34,825	0	0	0	0	
Server software	10,945	0	0	0	0	
Development labor	1,236,525	0	0	0	0	
TOTAL DEVELOPMENT COSTS:	1,632,295	0	0	0	0	
Hardware	54,000	81,261	81,261	81,261	81,261	
Software	20,000	20,000	20,000	20,000	20,000	
Operational labor	111,788	116,260	120,910	125,746	130,776	
TOTAL OPERATIONAL COSTS:	185,788	217,521	222,171	227,007	232,037	
TOTAL COSTS:	1,818,083	217,521	222,171	227,007	232,037	
PV OF COSTS:	1,765,129	205,034	203,318	201,693	200,157	2,575,331
PV OF ALL COSTS:	1,765,129	1,970,163	2,173,481	2,375,174	2,575,331	
TOTAL PROJECT BENEFITS COSTS:	(1,180,083)	450,479	477,629	506,501	537,201	
YEARLY NPV:	(1,145,712)	424,620	437,098	450,019	463,395	629,421
CUMULATIVE NPV:	(1,145,712)	(721,091)	(283,993)	166,026	629,421	
RETURN ON INVESTMENT:	24.44%	(629,421/2,575,331)				
BREAK-EVEN POINT:	3.63 years	[break-even occurs in year 4; (450,019 – 166,026)/450,019 = 0.63]				
INTANGIBLE BENEFITS:	This service is currently provided by competitors Improved customer satisfaction					

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Steps for Conducting Economic Feasibility

4-5. Determining Net Present Value and Return on Investment

Calculation	Definition	Formula
Present Value (PV)	The amount of an investment today compared to that same amount in the future, taking into account inflation and time.	$\frac{\text{Amount}}{(1 + \text{interest rate})^n}$ n = number of years in future
Net Present Value (NPV)	The present value of benefit less the present value of costs.	$\text{PV Benefits} - \text{PV Costs}$
Return on Investment (ROI)	The amount of revenues or cost savings results from a given investment.	$\frac{\text{Total benefits} - \text{Total costs}}{\text{Total costs}}$
Break-Even Point	The point in time at which the costs of the project equal the value it has delivered.	$\frac{\text{Yearly NPV}^* - \text{Cumulative NPV}}{\text{Yearly NPV}^*}$

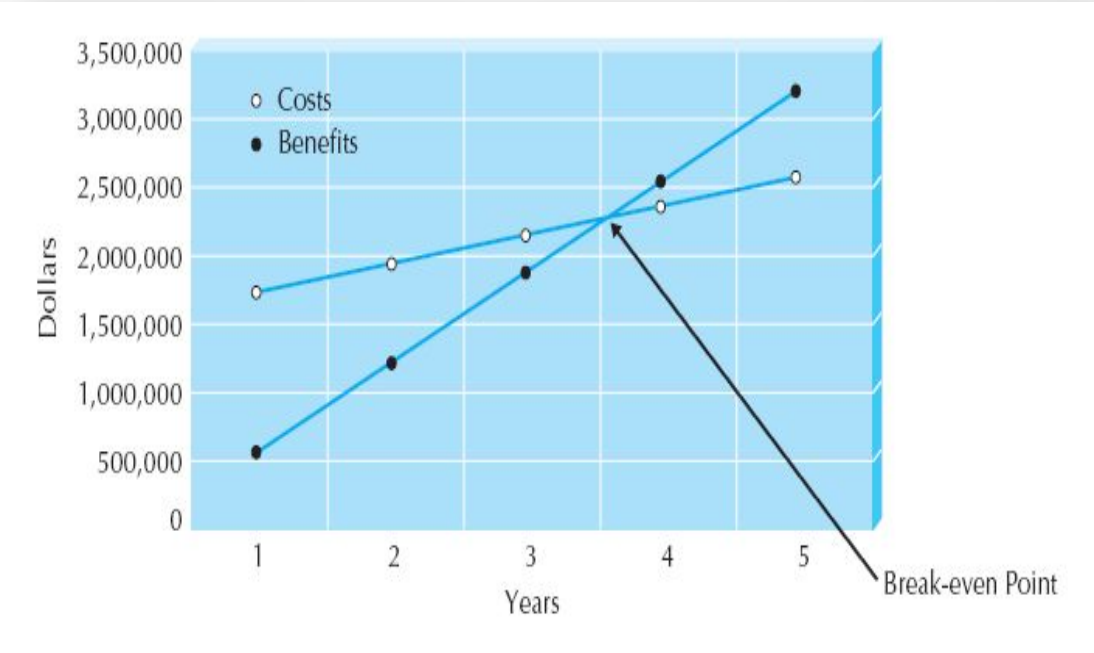
*Use the Yearly NPV amount from the first year in which the project has a positive cash flow.
Add the above amount to the year in which the project has a positive cash flow.

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Steps for Conducting Economic Feasibility

6-7 Break-Even Point



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Steps for Conducting Economic Feasibility

C. Organizational Feasibility (If we build it, will they come??)

Stakeholders

- ✓ Project champion(s)
- ✓ Senior management
- ✓ Users
- ✓ Others

Is the project strategically aligned with the business?

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Role		Techniques for Improvement
Champion	A champion: • Initiates the project • Promotes the project • Allocates his or her time to project • Provides resources	• Make a presentation about the objectives of the project and the proposed benefits to those executives who will benefit directly from the system • Create a prototype of the system to demonstrate its potential value
Organizational Management	Organizational managers: • Know about the project • Budget enough money for the project • Encourage users to accept and use the system	• Make a presentation to management about the objectives of the project and the proposed benefits • Market the benefits of the system using memos and organizational newsletters • Encourage the champion to talk about the project with his or her peers
System Users	Users: • Make decisions that influence the project • Perform hands-on activities for the project • Ultimately determine whether the project is successful by using or not using the system	• Assign users official roles on the project team • Assign users specific tasks to perform with clear deadlines • Ask for regular feedback from users (e.g., at weekly meetings)

Project Selection

Project portfolio management

- ✓ *A process that optimizes project selection and sequencing in order to best support business goals*

Business goals are expressed in terms of

- ✓ *Quantitative economic measures*
- ✓ *Business strategy goals*
- ✓ *IT strategy goals*

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Once selected, projects enter the project management process

Ways to Classify Projects

Size	What is the size? How many people are needed to work on the project?
Cost	How much will the project cost the organization?
Purpose	What is the purpose of the project? Is it meant to improve the technical infrastructure? Support a current business strategy? Improve operations? Demonstrate a new innovation?
Length	How long will the project take before completion? How much time will go by before value is delivered to the business?
Risk	How likely is it that the project will succeed or fail?
Scope	How much of the organization is affected by the system? A department? A division? The entire corporation?
Return on investment	How much money does the organization expect to receive in return for the amount the project costs?





TRADITIONAL PROJECT MANAGEMENT TOOLS

Project Management tools - used to successfully manage traditional software development projects (and many other types of projects)

1. *Work-breakdown structure*
2. *Gantt chart*
3. *network diagram*

Task

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is a unit of work that will be performed by a member or members of the development team, such as feasibility analysis. Each task is described by information such as its name, start and completion dates, person assigned to complete the task, deliverables, completion status, priority, resources needed, estimated time to complete the task, and the actual time it took to complete the task.



Task Information

Workplan Information	Example
Name of the task	Perform economic feasibility
Start date	Jan 05, 2015
Completion date	Jan 19, 2015
Person assigned to the task	Project sponsor: Mary Smith
Deliverable(s)	Cost-benefit analysis
Completion status	Open
Priority	High
Resources that are needed	Spreadsheet software
Estimated time	16 hours
Actual time	14.5 hours

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1. Work Breakdown Structures

A project manager can use a structured, top-down approach whereby high-level tasks are first defined and then broken down into subtasks.

There are two basic approaches to organizing a traditional WBS: by development phase or by product.

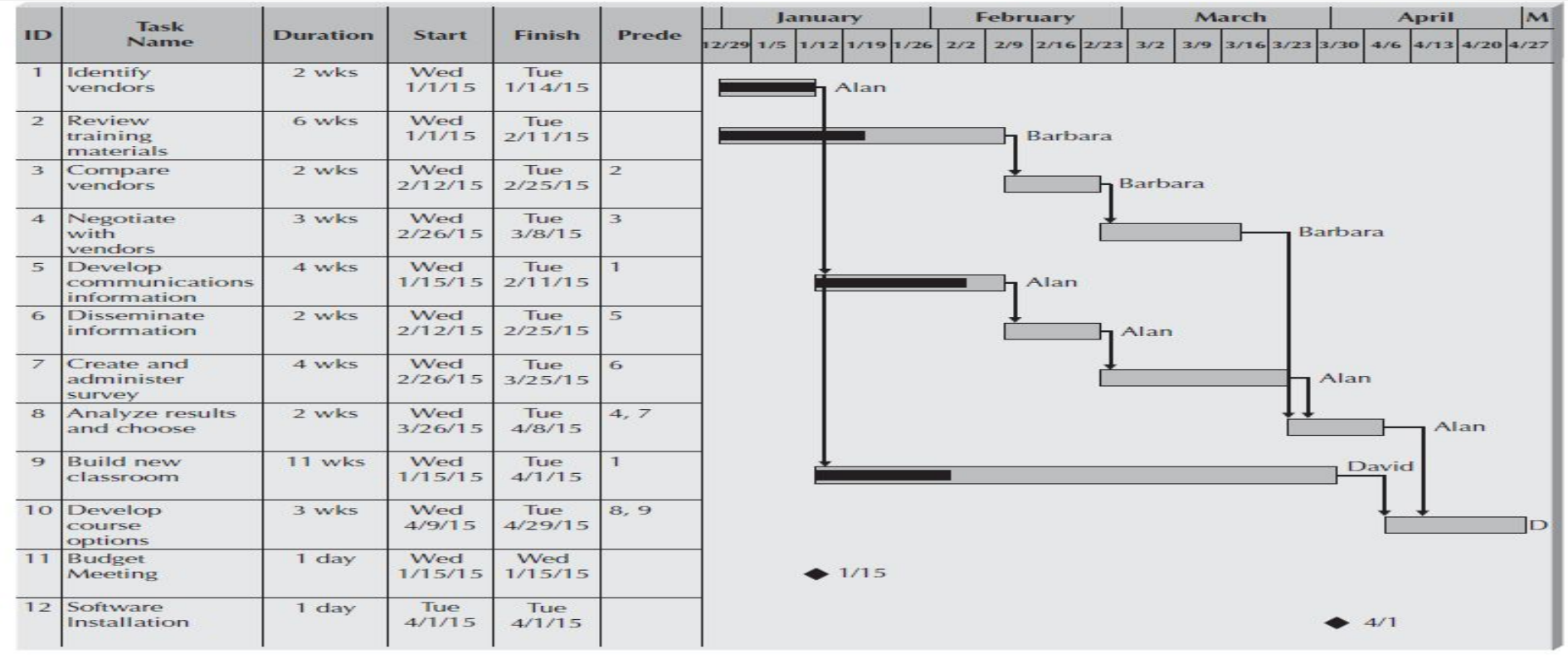
Task Number	Task Name	Duration (in weeks)	Dependency	Status
1	Identify vendors	2		Complete
2	Review training materials	6	1	Complete
3	Compare vendors	2	2	In Progress
4	Negotiate with vendors	3	3	Open
5	Develop communications information	4	1	In Progress
6	Disseminate information	2	5	Open
7	Create and administer survey	4	6	Open
7.1	Create initial survey	1		Open
7.2	Review initial survey	1	7.1	Open
7.2.1	Review by Director of IT Training	1		Open
7.2.2	Review by Project Sponsor	1		Open
7.2.3	Review by Representative Trainee	1		Open
7.3	Pilot test initial survey	1	7.1	Open
7.4	Incorporate survey changes	1	7.2, 7.3	Open
7.5	Create distribution list	0.5		Open
7.6	Send survey to distribution list	0.5	7.4, 7.5	Open
7.7	Send follow-up message	0.5	7.6	Open
7.8	Collect completed surveys	1	7.6	Open
8	Analyze results and choose vendor	2	4, 7	Open
9	Build new classrooms	11	1	In Progress
10	Develop course options	3	8, 9	Open

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2. Gantt Chart

A Gantt chart is a horizontal bar chart that shows the same task information as the project WBS but in a graphical way.



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3. Network Diagram

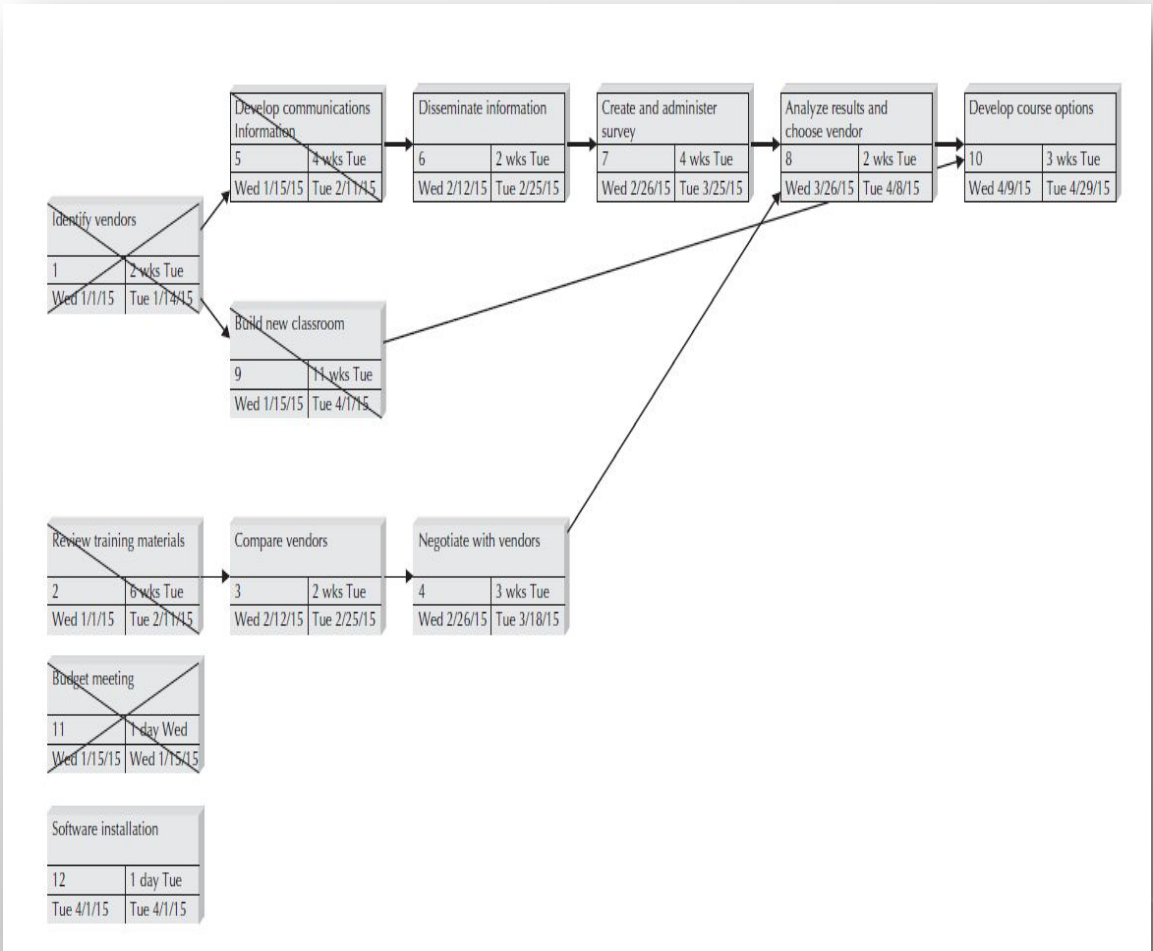
A second graphical way to look at project workplan information is the network diagram that lays out the project tasks in a flowchart.

The network diagram is drawn as a node-and-arc type of graph that shows time estimates in the nodes and task dependencies on the arcs. Each node represents an individual task, and a line connecting two nodes represents the dependency between two tasks. Partially completed tasks are usually displayed with a diagonal line through the node, and completed tasks contain crossed lines.

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The critical path method (CPM) simply allows the identification of the critical path in the network. The critical path is the longest path from the project inception to completion. The critical path shows all the tasks that must be completed on schedule for a project as a whole to finish on schedule.



Program Evaluation and Review Technique

(PERT) is a network analysis technique that can be used when the individual task time estimates are fairly uncertain. Instead of simply putting a point estimate for the duration estimate, PERT uses three-time estimates: optimistic, most likely, and a pessimistic. It then combines the three estimates into a single weighted average estimate using the following form

$$\text{PERT weighted average} = \frac{\text{optimistic estimate} + (4 * \text{most likely estimate}) + \text{pessimistic estimate}}{6}$$

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Project Estimation

The process of assigning projected values for time and effort. The estimates developed at the start of a project are usually based on a range of possible values and gradually become more specific as the project moves forward. That is, the range of values for the inception phase will be much greater than for the transition phase.

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ENVIRONMENT INFRASTRUCTURE MANAGEMENT

AND

The environment and infrastructure management workflows support the development team throughout the development process. The environment workflow primarily deals with choosing the correct set of tools that will be used throughout the development process and identifying the appropriate set of standards to be followed during the development process.

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CASE Tools

Computer-aided software engineering (CASE) is a category of software that automates all or part of the development process. Some CASE software packages are used primarily to support the analysis workflow to create integrated diagrams of the system and to store information regarding the system components, whereas others support the design workflow that can be used to generate code for database tables and system functionality.

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Standards

One way to make certain that everyone is performing tasks in the same way and following the same procedures are to create standards that the project team must follow. Standards can include formal rules for naming files, forms that must be completed when goals are reached, and programming guidelines

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Standards

Types of Standards	Examples
Documentation standards	- The date and project name should appear as a header on all documentation. - All margins should be set to 1 inch. - All deliverables should be added to the project binder and recorded in its table of contents.
Coding standards	- All modules of code should include a header that lists the programmer, last date of update, and a short description of the purpose of the code. - Indentation should be used to indicate loops, if-then-else statements, and case statements. - On average, every program should include one line of comments for every five lines of code.
Procedural standards	- Record actual task progress in the work plan every Monday morning by 10 AM. - Report to project update meeting on Fridays at 3:30 PM. - All changes to a requirements document must be approved by the project manager.
Specification requirement standards	- Name of program to be created - Description of the program's purpose - Special calculations that need to be computed - Business rules that must be incorporated into the program - Pseudocode - Due date
User interface design standards	- Labels will appear in boldface text, left-justified, and followed by a colon. - The tab order of the screen will move from top left to bottom right. - Accelerator keys will be provided for all updatable fields.

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Documentation

The documentation typically includes the system request, the feasibility analysis, the original and later versions of the effort estimation, the evolving workplan, and UML diagrams for the functional, structural, and behavioral models. A poor project management practice is waiting until the last minute to create documentation.

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